

PIARCO INTL, Trinidad and Tobago

WMO: 789700

Lat: 10.595N

Lon: 61.337W

Elev: 12

StdP: 101.18

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 11634

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	21.8	18.1	13.0	28.9	18.9	13.7	28.1	9.4	29.6	8.5	29.5	0.1	90	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.7	34.0	25.8	33.2	25.5	33.0	25.4	27.2	31.2	26.8	30.9	26.6	30.6	5.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.6	29.2	25.9	21.3	29.0	25.4	20.6	28.3	85.8	31.4	84.5	31.2	83.4	31.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	8.0	7.3	DB	19.5	35.4	0.8	1.0	19.0	36.1	18.5	36.7	18.1	37.2	17.5	37.9	(4)
(5)			WB	18.8	28.5	0.7	0.8	18.2	29.1	17.8	29.6	17.4	30.0	16.9	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.5	26.6	27.2	28.0	28.4	27.9	27.9	28.2	28.4	28.2	27.5	26.9		
	DBStd	1.12	0.74	0.94	1.00	1.01	0.82	0.92	0.88	0.90	0.90	1.04	0.94	0.86		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6439	511	465	534	540	569	536	554	563	553	563	526	524		
	CDD18.3	3397	253	232	276	290	311	286	296	305	303	304	276	266		
	CDH23.3	32561	2070	1940	2575	2964	3301	2842	2939	3063	3114	2998	2499	2256		
	CDH26.7	11987	709	689	1008	1194	1273	949	1046	1175	1231	1120	852	741		
Wind	WSAvg	2.7	2.8	3.3	3.4	3.5	3.6	3.3	2.6	1.9	1.8	1.9	2.1	2.5		
Precipitation	PrecAvg	1766	78	44	37	47	101	249	256	242	186	187	225	151		
	PrecMax	2192	247	164	135	216	260	478	387	449	328	488	443	331		
	PrecMin	1456	9	2	1	6	12	83	116	108	56	36	67	48		
	PrecStd	202	55	31	31	44	60	85	65	80	57	86	86	67		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.7	33.8	34.2	34.0	33.2	33.2	34.2	34.6	34.2	33.2	32.2		
		MCWB	24.0	24.2	24.5	25.0	24.7	25.3	25.7	26.2	26.2	26.3	25.9	25.1		
	2%	DB	31.2	31.8	32.9	33.2	33.1	32.2	32.9	33.2	33.8	33.2	32.4	31.8		
		MCWB	24.0	23.9	24.1	24.6	24.8	25.3	25.6	26.1	26.2	26.2	25.8	25.0		
	5%	DB	30.9	31.1	32.0	32.6	32.7	31.9	32.1	32.8	33.0	32.8	31.9	31.1		
		MCWB	24.0	23.7	24.0	24.4	24.8	25.3	25.5	26.0	26.1	26.1	25.8	25.0		
	10%	DB	30.1	30.2	31.1	31.9	32.0	31.1	31.2	31.9	32.1	31.9	31.0	30.2		
		MCWB	23.9	23.5	23.8	24.3	24.9	25.4	25.6	25.9	26.0	26.0	25.9	24.9		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	25.8	25.9	26.6	27.0	27.0	27.2	27.4	27.6	27.4	27.2	26.7		
		MCDB	29.4	30.0	29.8	30.7	30.4	30.4	30.7	31.7	31.7	31.7	30.9	30.2		
	2%	WB	25.5	25.0	25.3	26.0	26.5	26.6	26.7	26.9	27.0	27.0	26.7	26.2		
		MCDB	29.1	29.3	29.7	30.0	29.9	29.9	30.4	31.2	31.5	31.3	30.4	29.7		
	5%	WB	25.0	24.6	24.9	25.6	26.1	26.2	26.3	26.6	26.7	26.6	26.4	25.8		
		MCDB	28.9	28.8	29.4	29.6	29.5	29.6	30.1	30.8	31.0	30.8	30.1	29.2		
	10%	WB	24.6	24.2	24.5	25.2	25.7	25.9	26.0	26.3	26.4	26.3	26.1	25.4		
		MCDB	28.3	28.3	28.9	29.3	29.1	29.2	29.8	30.4	30.6	30.4	29.7	28.8		
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.4	8.8	8.5	7.7	6.8	7.3	8.1	8.3	8.0	7.6	7.7		
		MCDBR	9.0	9.1	9.5	9.2	8.6	7.7	8.3	9.1	9.2	9.2	8.5	8.5		
		MCWBR	3.2	2.9	2.8	2.4	2.3	2.5	2.9	3.2	3.2	3.3	3.2	3.2		
	5% WB	MCDBR	7.7	8.1	8.4	7.9	7.2	6.7	7.4	8.2	8.5	8.2	7.7	7.4		
		MCWBR	3.2	3.0	2.8	2.5	2.6	2.7	3.1	3.3	3.3	3.3	3.3	3.2		
Clear-Sky Solar Irradiance	taub	0.385	0.407	0.405	0.454	0.471	0.515	0.489	0.436	0.420	0.395	0.385	0.398			
	taud	2.533	2.439	2.456	2.316	2.270	2.123	2.219	2.417	2.451	2.550	2.575	2.520			
	Ebn at Noon	923	911	915	860	830	783	808	864	887	911	916	901			
	Edh at Noon	104	117	117	134	137	156	143	119	116	104	99	103			
All-Sky Solar Radiation	RadAvg	5.10	5.65	5.91	5.94	5.65	5.26	5.42	5.62	5.65	5.18	4.72	4.63			
	RadStd	0.28	0.25	0.33	0.27	0.40	0.25	0.28	0.24	0.25	0.22	0.31	0.28			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.25	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.20	+0.33	+0.41	+0.14	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page